Autocallable Notes on the CAC40 under a Calibrated Heston Model: Athena and Phoenix, Fair Coupons, Scenarios and Greeks

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Abstract

We price the two main families of single-underlying autocallable notes, the Athena (coupon paid at the call only) and the Phoenix (periodic conditional coupon, with or without memory), on the CAC40 of 12 February 2025. The underlying follows a Heston model calibrated to the implied volatility surface (RMSE 0.42 volatility point), with a drift read from a forward curve bootstrapped from put–call parity. Both products are described by a single parametrisation covering the observation frequency, snowball or fixed coupons, coupon barrier and memory, autocall barrier and step-down, non-call period, European, American continuous or American daily capital protection, airbag and capital guarantee. The notes are priced by Monte Carlo with antithetic variates and a control variate; the American protection is handled with the Brownian bridge crossing probability as a conditional expectation. The pricer recovers five closed-form limit cases within one standard error, and an exact identity between the two families. We report the fair coupon (price at par) of 35 variants together with their scenario statistics: the base Athena (4 years, annual observations, autocall at 100%, European protection at 60%) pays 9.41% a year, is called at the first date with probability 55.7% and loses capital with probability 9.1%; the base Phoenix (quarterly, memory, coupon barrier at 70%) pays 6.04% a year. Switching to an American protection adds 1.77 points of coupon to the Athena. Both notes are short volatility (-0.55 and -0.33 per vol point on a nominal of 100) and short dividends (-1.12 and -0.32 per point of carry).

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1 Introduction

Autocallable notes are the most widely distributed structured products on equity indices in Europe. The investor receives coupons well above the risk-free rate, and the note is redeemed early ("called") as soon as the index is above a reference level on an observation date. The coupon is financed by a short down-and-in put: if the note is never called and the index ends below a protection barrier, the investor bears the loss of the index.

Two families cover most of the market:

- the **Athena**, whose coupon is paid only when the note is called, usually accumulated over time (*snowball*);
- the **Phoenix**, whose coupon is paid at each observation date where the index is above a coupon barrier, lower than the autocall barrier, usually with a *memory* of the missed coupons.

On top of these, issuers combine many features: observation frequency, step-down of the autocall barrier, non-call period, European or American protection, airbag, capital guarantee. This report prices all of them in a single framework, on real market data, and measures how each feature moves the fair coupon, the probabilities of the scenarios and the Greeks.

The model is the Heston model calibrated in [1] to the CAC40 implied volatility surface, and the Monte Carlo engine is the one used for barrier options in [2]. The scripts and notebooks are open source; Section 8 describes the code.

2 Products

2.1 Common structure

A note of nominal N on an index S has an initial fixing S_{ref} and observation dates $0 < t_1 < \cdots < t_n = T$. The performance at t_k is $P_k = S_{t_k}/S_{\text{ref}}$. All barriers are fractions of S_{ref}: autocall barriers B_k^{call}, coupon barriers B_k^{cpn} (Phoenix), protection barrier B^{prot}. The dates t_k with $t_k \geq t_{\text{nc}}$ are callable, t_{nc} being the end of the non-call period; the maturity is always callable. The first callable date where $P_k \geq B_k^{\text{call}}$ is the call date τ.

Redemption at maturity. If the note is never called, it is redeemed at T at

$$R_T = N\left[(1 - \mathbf{1}_{\text{KI}}) + \mathbf{1}_{\text{KI}} \max\Big(\phi,\ \min\left(1,\ P_T/K_{\text{put}}\right)\Big)\right], \tag{1}$$

where $\mathbf{1}_{\text{KI}}$ is the knock-in event of the protection, K_{put} the strike of the short put and ϕ the capital floor. The knock-in event is

- *European protection*: $P_T < B^{\text{prot}}$;
- *American protection*: the index touches $B^{\text{prot}} S_{\text{ref}}$ at any time before maturity, continuously or on daily closes.

The standard note has $K_{\text{put}} = 1$ and $\phi = 0$: below the barrier, the investor receives NP_T and the redemption jumps from N to NB^{prot} at the barrier. The *airbag* note has $K_{\text{put}} = B^{\text{prot}}$: the redemption is continuous at the barrier and equal to NP_T/B^{prot} below it. A *capital guarantee* sets $\phi > 0$. Figure 1 draws the three profiles.

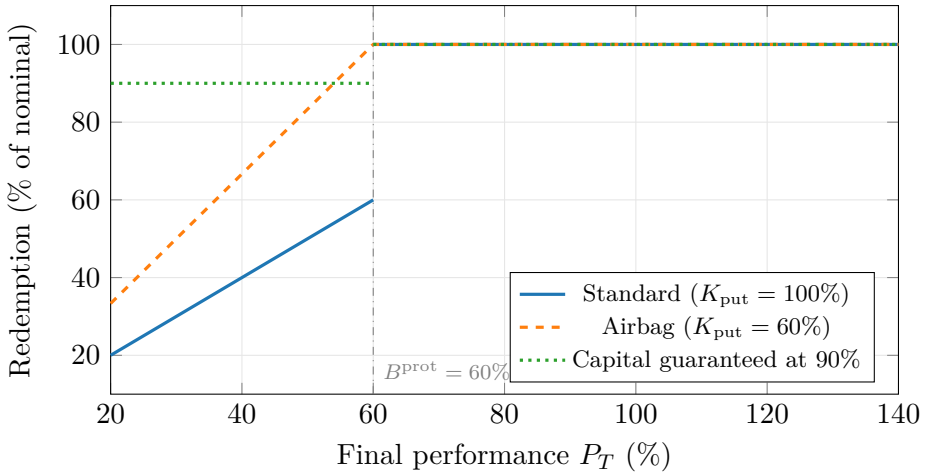


Figure 1: Redemption at maturity of a note that was never called, European protection at 60%.

2.2 Athena

The Athena pays a coupon only with the redemption at the call date. If $\tau = t_k$, the investor receives

$$N(1 + c_k), \qquad c_k = \begin{cases} c\, t_k & \text{snowball (the coupons accumulate),} \\ c\,(t_k - t_{k-1}) & \text{fixed coupon,} \end{cases} \tag{2}$$

with c the coupon rate per annum. If the note is not called, no coupon is paid and the redemption is (1). A call at maturity ($P_T \geq B_n^{\text{call}}$) pays the last coupon.

2.3 Phoenix

The Phoenix pays at each observation date $t_k \leq \tau$ (and $t_k \leq T$) the coupon

$$N\, c\,(t_k - t_{k-1})\, \mathbf{1}_{\{P_k \geq B_k^{\text{cpn}}\}} \tag{3}$$

and, with memory, the coupons of all the previous dates since the last payment. When the note is called, it is redeemed at N. With $B_k^{\text{cpn}} = 0$, the coupon is unconditional; with $B_k^{\text{cpn}} = B_k^{\text{call}}$ and memory, a coupon is paid only at the call date and its amount is $c\, t_k$: the Phoenix is then exactly a snowball Athena (Table 3 checks it numerically). Figure 2 summarises the decisions taken at an observation date.

2.4 Parametrisation

Table 1 lists the parameters of the classes `Athena` and `Phoenix`; every variant of Sections 5 and 6 is one of these parameters changed.

3 Market Data and Model

The data are the CAC40 option chain of 12 February 2025 (spot $S_0 = 8042.19$, 142 quotes on 13 expiries from 9 days to 4.86 years) and the EURIBOR 6M zero-coupon curve. The forward curve $F(0,t)$ is bootstrapped from put–call parity with a piecewise constant borrow-plus-dividend yield $q(t)$ [1]; its integral reaches 0.136 at 4.86 years, so dividends lower the 4-year forward by about 11% with respect to a zero carry. Under $\mathbb{Q}$,

$$\frac{\mathrm{d}S_t}{S_t} = \partial_t \ln F(0,t)\,\mathrm{d}t + \sqrt{v_t}\,\mathrm{d}W_t^S, \qquad \mathrm{d}v_t = \kappa(\theta - v_t)\,\mathrm{d}t + \xi\sqrt{v_t}\,\mathrm{d}W_t^v, \qquad \mathrm{d}\langle W^S, W^v\rangle_t = \rho\,\mathrm{d}t, \tag{4}$$

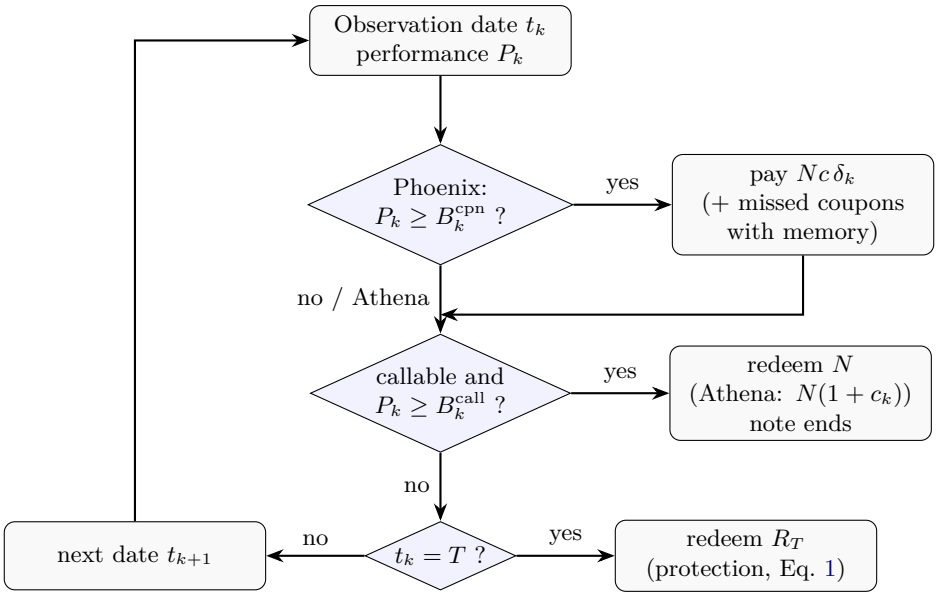


Figure 2: Cash flows of an autocallable note at an observation date.

Feature	**Parameter**	**Values**
Maturity, initial fixing, nominal	`maturity`, `initial_fixing`, `nominal`	years, index level, 100
Observation dates	`frequency` or `observation_dates`	monthly to annual, or any list ending at T
Coupon rate	`coupon_rate`	per annum, fraction of the nominal
Athena coupon	`coupon_type`	`snowball`, `fixed`
Phoenix coupon	`coupon_barrier`, `memory`	fraction of S_{ref} (scalar or one per date), True / False
Autocall barrier	`autocall_barrier`, `step_down`	fraction of S_{ref}, decrease per observation
Non-call period	`first_call_date`	years
Capital protection	`protection_barrier`, `protection_type`	fraction of S_{ref}; `european`, `american`, `american_discrete`
Airbag, guarantee	`put_strike`, `floor`	fraction of S_{ref}, fraction of the nominal

Table 1: Parameters of the autocallable notes (`Produits/products.py`).

with the parameters of Table 2. The notes are struck at the money: $S_{\text{ref}} = S_0 = 8042.19$, and S_{ref} stays fixed when the Greeks bump the market. All the dates of the 4-year notes lie before the last listed expiry (4.86 years).

v_0	θ	κ	ξ	ρ	IV RMSE
0.0197	0.0438	1.308	0.482	-0.605	0.421 vol pts

Table 2: Heston parameters calibrated on the CAC40 surface of 12 February 2025 [1].

4 Pricing Method

4.1 Simulation

The paths of (S, v) are simulated with an Euler full truncation scheme on a weekly grid (52 steps a year, 208 steps for 4 years), the drift over each step being $\ln(F(0,t_{i+1})/F(0,t_i))$, so that $\mathbb{E}[S_t] = F(0,t)$ at every date. $N = 50{,}000$ paths with antithetic variates are used (seed 42); prices and standard errors are computed on the 25,000 independent pair averages. The observation dates are rounded to the grid; with weekly steps, every monthly, quarterly, semi-annual and annual date falls within half a week of its contractual date.

4.2 Cash flows and protection

For each path, the note is followed through its observation dates (Figure 2), each cash flow being discounted with $D(0,t_k) = e^{-r(t_k)t_k}$. The knock-in indicator of (1) is

- *European*: $\mathbf{1}_{\{P_T < B^{\text{prot}}\}}$;
- *American continuous*: replaced by its conditional probability given the simulated path, $1 - \prod_i (1-p_i)$, where $p_i = \exp(-2\ln(S_{t_i}/H)\ln(S_{t_{i+1}}/H)/(v_{t_i}\Delta t))$ is the probability that the Brownian bridge of $\ln S$ touches $H = B^{\text{prot}} S_{\text{ref}}$ between two grid dates [7];
- *American daily*: the barrier is checked on a daily grid (252 steps a year).

4.3 Variance reduction

Besides antithetic variates, the discounted European put struck at $B^{\text{prot}} S_{\text{ref}}$, whose Heston price is known semi-analytically (Lewis formula), is used as a control variate with an optimal coefficient estimated on the paths.

4.4 Fair coupon and scenario statistics

On a fixed set of paths, the call dates do not depend on the coupon, and every cash flow is affine in the coupon rate c: the price is $V(c) = V(0) + c\,(V(1) - V(0))$, and the coupon that prices the note at par is

$$c^\star = \frac{N - V(0)}{V(1) - V(0)}, \tag{5}$$

obtained with two valuations. It excludes the issuer margin and funding spread. The same paths give the probability of a call at each date, the probability to reach maturity, the probability of a capital loss ($\mathbf{1}_{\text{KI}} = 1$ and $P_T < K_{\text{put}}$), the expected loss at maturity, the expected life $\mathbb{E}[\min(\tau, T)]$ and the expected number of coupons.

4.5 Greeks

The Greeks are central finite differences with common random numbers [6], on a nominal of 100: Delta and Gamma (spot $\pm 1\%$, carry kept), Vega and Volga ($\sqrt{v_0}$ and $\sqrt{\theta}$ shifted by ± 1 vol point), Vanna, Rho (± 10bp on the zero curve, per $+1\%$), Dividend (± 10bp on $q(t)$, per $+1\%$) and Theta (dates moved by one day). Delta is given per index point: multiply by $S_0/100 = 80.4$ for the value change of a 1% move. Each Greek is reported with its Monte Carlo standard error.

4.6 Validation

Five limit cases have a closed form or an exact counterpart (Table 3). In a flat Black–Scholes market ($S_0 = S_{\text{ref}} = 100$, $r = 3\%$, $q = 1\%$, $\sigma = 20\%$, $T = 2$, $N = 100$, Heston with $v_0 = \theta = \sigma^2$ and $\xi = 10^{-4}$, 10^5 paths):

- a note never called (autocall barrier out of reach) with a European protection at $B = 60$ is worth $ND(T) - \frac{N}{S_{\text{ref}}}\left[\text{Put}(B) + (S_{\text{ref}} - B)\,\text{DigitalPut}(B)\right]$;
- with an American protection, $ND(T) - \frac{N}{S_{\text{ref}}}\,\text{PDI}(K = S_{\text{ref}}, H = B)$ (Reiner–Rubinstein);
- a note called for sure at t_1 is worth $N(1 + c\,t_1)D(t_1)$;
- a Phoenix never called with unconditional quarterly coupons is worth the sum of its discounted coupons plus the first case.

On the CAC40, the Phoenix with memory and $B^{\text{cpn}} = B^{\text{call}}$ must have exactly the price of the snowball Athena with the same dates.

Case	Reference	Monte Carlo	Std error	Error / std error
Athena never called, European protection (flat)	92.6369	92.6480	0.0141	0.79
Athena never called, American protection (flat)	91.5500	91.5483	0.0220	−0.07
Athena called for sure at $t_1 = 0.5$, $c = 8\%$ (flat)	102.4516	102.4516	0	exact
Phoenix never called, unconditional 6% coupon (flat)	104.2404	104.2515	0.0141	0.79
Phoenix called for sure at $t_1 = 0.25$, $c = 6\%$ (flat)	100.742	100.742	0	exact
Phoenix with memory, $B^{\text{cpn}} = B^{\text{call}}$ vs snowball Athena (CAC40)	98.7669	98.7669	0.0585	identical paths

Table 3: Validation of the pricer.

5 Athena

5.1 Base case

The base Athena has a maturity of 4 years, annual observations, a snowball coupon of 8% a year, an autocall barrier at 100% and a European protection at 60%. Its price is 98.596 ± 0.128 (95%) for a nominal of 100, and its fair coupon is 9.41% a year. Table 4 gives the scenario statistics and Figure 3 the call probabilities: more than half of the notes are called after one year, a quarter reaches maturity, and 9.1% lose capital, with an expected loss of 5.27% of the nominal (58% of the nominal given a loss).

Call at 1 year	55.69%	Probability to reach maturity	25.70%
Call at 2 years	10.83%	Probability of a capital loss	9.06%
Call at 3 years	4.81%	Expected loss at maturity	5.27%
Call at 4 years	2.96%	Expected life	2.06 years

Table 4: Base Athena: scenario statistics on 50,000 paths.

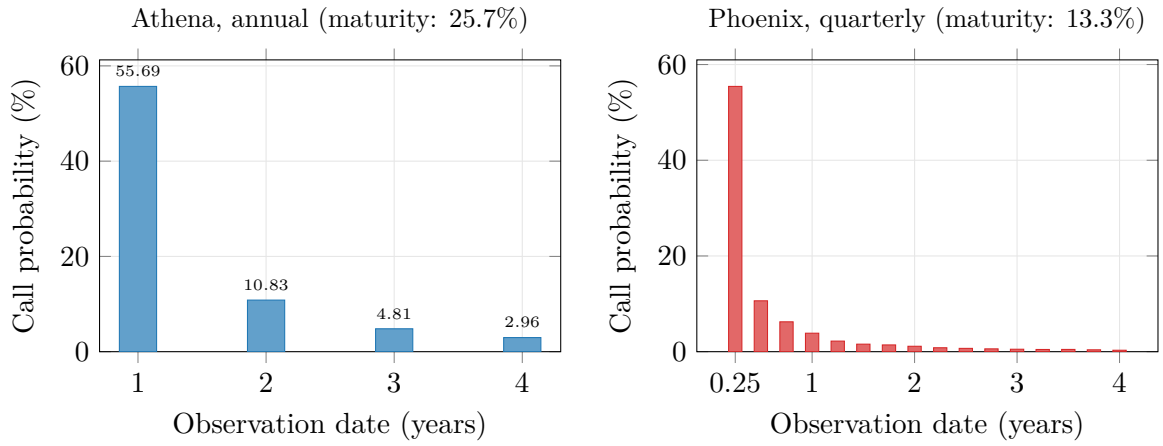


Figure 3: Probability of being called at each observation date (autocall barrier at 100%).

5.2 All the forms of the Athena

Table 5 changes one feature of the base case at a time. All prices are computed at the same coupon of 8% (standard errors between 0.03 and 0.08); the fair coupon is the comparable quantity.

Reading the table. The investor is short a down-and-in put whose premium is paid back through the coupons; the fair coupon per annum is roughly this premium divided by the expected life of the note.

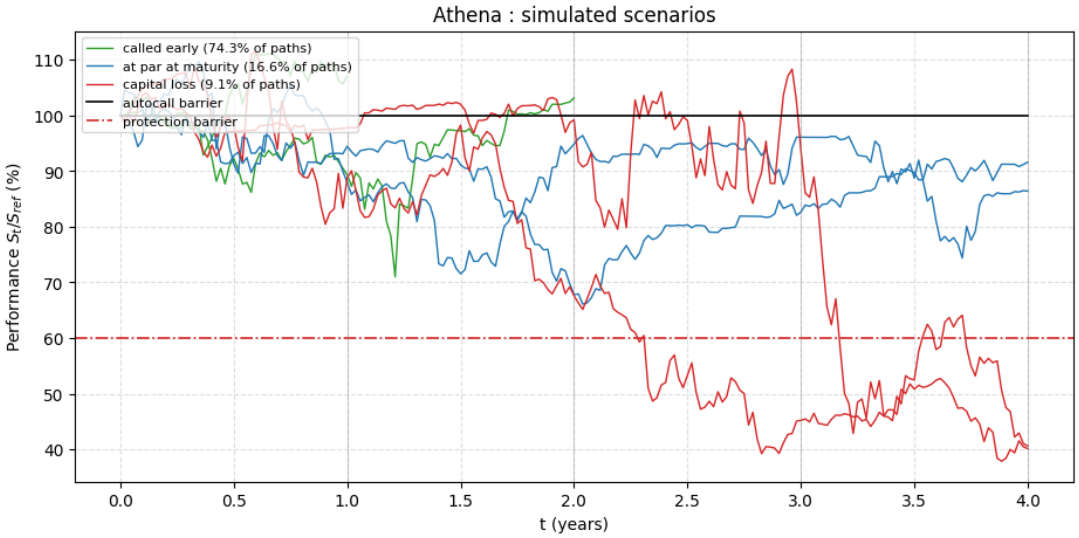


Figure 4: Base Athena: simulated paths of the performance, grouped by outcome.

Variant	Price at 8%	Fair coupon (% p.a.)	P(call t_1) (%)	P(maturity) (%)	P(loss) (%)	E[loss] (%)	E[life] (y)
Base (annual, snowball, AC 100%, EU 60%)	98.60	9.41	55.69	25.70	9.06	5.27	2.06
Semi-annual observations	98.64	9.93	53.30	18.61	6.97	4.08	1.48
Quarterly observations	98.77	10.54	55.43	13.29	5.22	3.05	1.03
Monthly observations	99.16	11.11	56.21	7.77	3.13	1.83	0.59
Fixed coupon (no snowball)	96.40	13.00	55.69	25.70	9.06	5.27	2.06
Autocall barrier 90%	100.95	7.03	77.37	13.24	6.41	3.77	1.54
Autocall barrier 110%	95.40	12.71	25.07	45.60	11.77	6.77	2.85
Step-down 5% per year	100.65	7.46	55.69	17.56	8.57	5.01	1.95
Quarterly, non-call 1 year	99.09	8.94	0.00	22.06	8.42	4.92	1.89
Protection 50%	99.90	8.10	55.69	25.70	5.91	3.86	2.06
Protection 70%	97.33	10.68	55.69	25.70	12.96	6.63	2.06
American protection (continuous)	96.83	11.18	55.69	25.70	16.43	7.20	2.06
American protection (daily)	96.95	11.05	55.53	25.67	16.07	7.18	2.07
Airbag	100.89	7.10	55.69	25.70	9.06	2.75	2.06
Capital guaranteed 90%	102.56	5.42	55.69	25.70	9.06	0.91	2.06
Capital guaranteed 100%	103.39	4.59	55.69	25.70	0.00	0.00	2.06

Table 5: Athena variants on the CAC40 (Heston, 50,000 paths). E[loss] in % of the nominal.

- *Observation frequency.* More frequent observations call the note earlier (expected life from 2.06 to 0.59 years) and reduce the probability of a loss (from 9.06% to 3.13%), but the put premium is spread over a much shorter life: the fair coupon rises from 9.41% to 11.11%.
- *Snowball vs fixed coupon.* A fixed coupon pays c once whatever the call date, instead of $c\,t_k$: the fair coupon is 13.00% instead of 9.41%.
- *Autocall barrier.* A barrier at 90% calls 77% of the notes after one year and lowers the fair coupon to 7.03%; at 110%, 46% of the notes reach maturity and the fair coupon is 12.71%. A step-down of 5% a year lowers it to 7.46%.
- *Protection.* Each 5 points of protection barrier move the fair coupon by 0.4 to 0.7 point (Figure 5). The American protection raises the probability of a loss from 9.06% to 16.43% and the fair coupon by 1.77 points; a daily monitoring lowers the fair coupon by 0.13 point with respect to a continuous monitoring.
- *Airbag and guarantee.* The airbag halves the expected loss (2.75% instead of 5.27%) for a coupon of 7.10%; a 100% capital guarantee leaves a coupon of 4.59%, the value of the call-at-par feature on top of a zero-coupon bond.

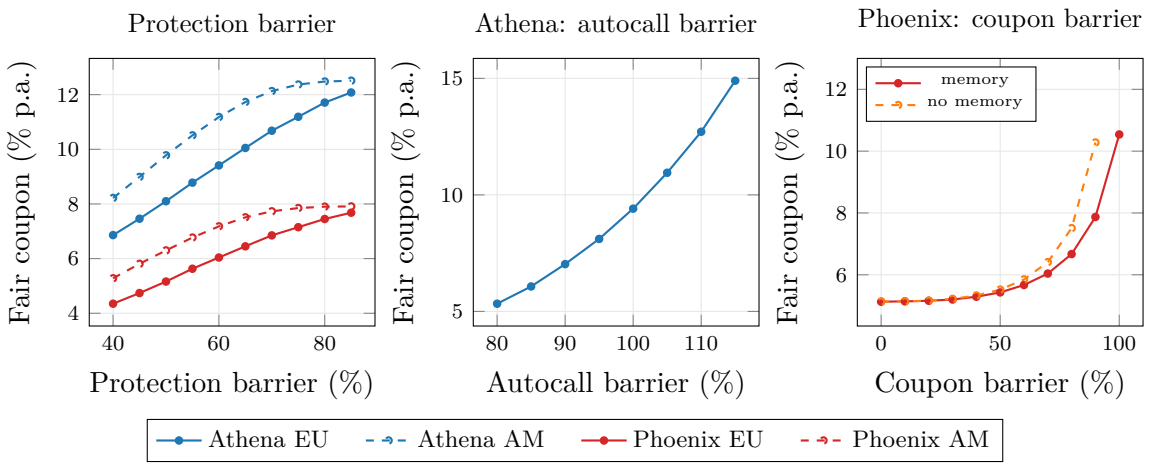


Figure 5: Fair coupons as functions of the barriers (other features of the base cases unchanged; EU / AM: European / American continuous protection). The Phoenix without memory reaches 23.94% at a coupon barrier of 100% (off scale).

5.3 Greeks

Table 6 gives the Greeks of five Athena variants. Gamma and Volga are not significant at this number of paths (base case: Gamma $-3.0 \times 10^{-6} \pm 5.4 \times 10^{-6}$, Volga -0.042 ± 0.037); the other Greeks have relative standard errors between 3% and 45% (base case: Delta 0.00506 ± 0.00019, Vega -0.549 ± 0.016, Rho -0.77 ± 0.07, Dividend -1.12 ± 0.07, Theta 0.0050 ± 0.0016).

Athena	**Price**	**Delta** (per point)	**Delta 1%** (per 1% move)	**Vega** (per vol pt)	**Vanna** ($\times 10^4$)	**Rho** (per 1%)	**Dividend** (per 1%)	**Theta** (per day)
Base	98.60	0.00506	0.407	−0.549	2.51	−0.770	−1.115	0.0050
American protection	96.83	0.00644	0.518	−0.657	1.97	−0.546	−1.269	0.0050
Step-down 5%	100.65	0.00321	0.258	−0.637	6.36	−0.900	−0.935	0.0079
Airbag	100.89	0.00343	0.276	−0.354	1.28	−1.161	−0.815	0.0037
Guaranteed 90%	102.56	0.00241	0.193	−0.139	0.44	−1.436	−0.608	0.0023

Table 6: Greeks of Athena variants (nominal 100, coupon 8%).

The note is long the index (a 1% rise is worth 0.41 on 100) and short volatility: one vol point costs 0.55, the short put dominating the long digital-like call features. It is also short dividends: one point of carry lowers every forward and costs 1.12, twice the vega. It loses with higher rates (discounting of the coupons and of the nominal) and earns about 0.005 a day (1.8 a year) as the put decays. The capital guarantee reduces every risk but the rate risk, since the guaranteed redemption behaves as a zero-coupon bond. Figure 6 displays the table with 95% confidence intervals and Figure 7 the Greeks as functions of the spot, the initial fixing staying at 8042: the Delta increases as the spot falls towards the protection barrier (4825), and peaks just above it with an American protection.

Sensitivities to the Heston parameters. Per unit of parameter, the base Athena has $\partial V/\partial v_0 = -69.9$, $\partial V/\partial\theta = -69.4$, $\partial V/\partial\kappa = -0.81$, $\partial V/\partial\xi = 3.94$ and $\partial V/\partial\rho = -1.03$. The note loses with the variance level (short put) and gains with the vol-of-vol ξ.

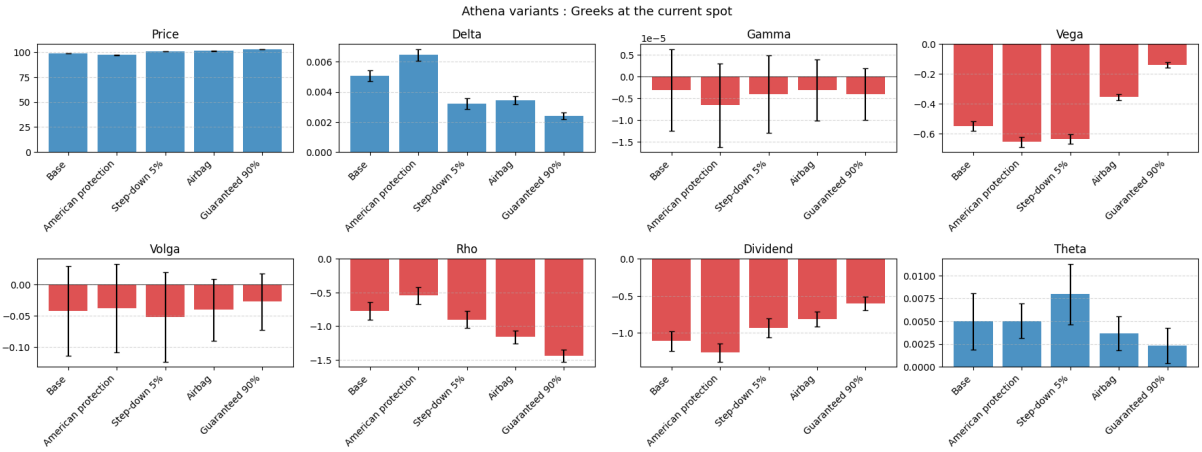


Figure 6: Greeks of the Athena variants with 95% confidence intervals.

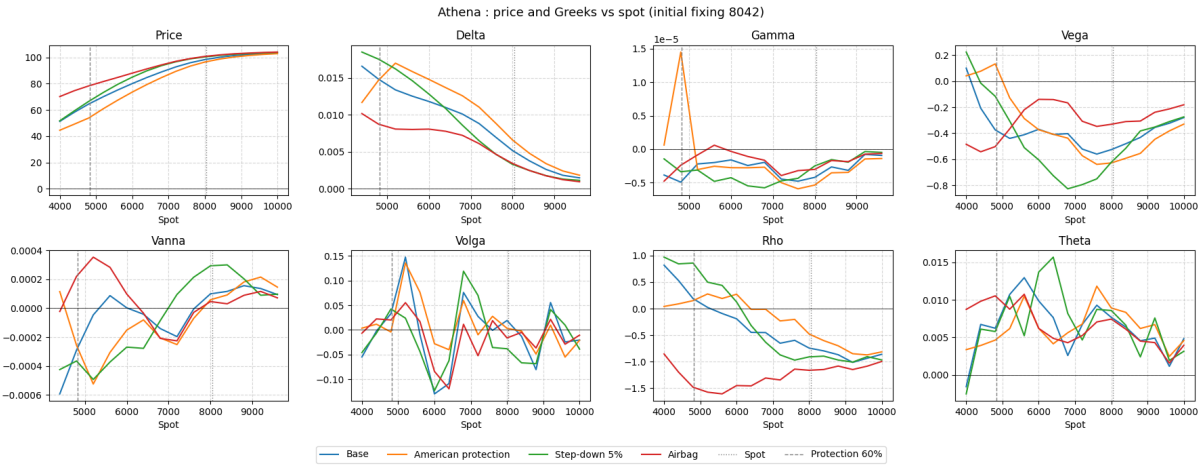


Figure 7: Athena: price and Greeks as functions of the spot (30,000 paths, spot step 400).

6 Phoenix

6.1 Base case

The base Phoenix has a maturity of 4 years, quarterly observations, a coupon of 6% a year (1.5% a quarter) with memory and a coupon barrier at 70%, an autocall barrier at 100% and a European protection at 60%. Its price is 99.963 ± 0.099 (95%) and its fair coupon 6.04% a year. 55.4% of the notes are called after one quarter (Figure 3), the expected life is 1.03 years, 3.28 coupons are paid on average, 13.3% of the notes reach maturity and 5.2% lose capital, with an expected loss of 3.05%.

6.2 All the forms of the Phoenix

Reading the table.

- *Coupon barrier and memory.* Lowering the coupon barrier from 80% to 50% raises the expected number of coupons from 2.80 to 3.83 and lowers the fair coupon from 6.67% to 5.43%; an unconditional coupon gives 5.13%. Memory is worth 0.37 point of coupon at a barrier of 70% (6.04% instead of 6.41%), and its value grows with the barrier (Figure 5): at 90%, 7.87% with memory against 10.28% without. With the coupon barrier equal to the autocall barrier, the memory Phoenix is the snowball Athena (10.54%, as the quarterly Athena of Table 5).

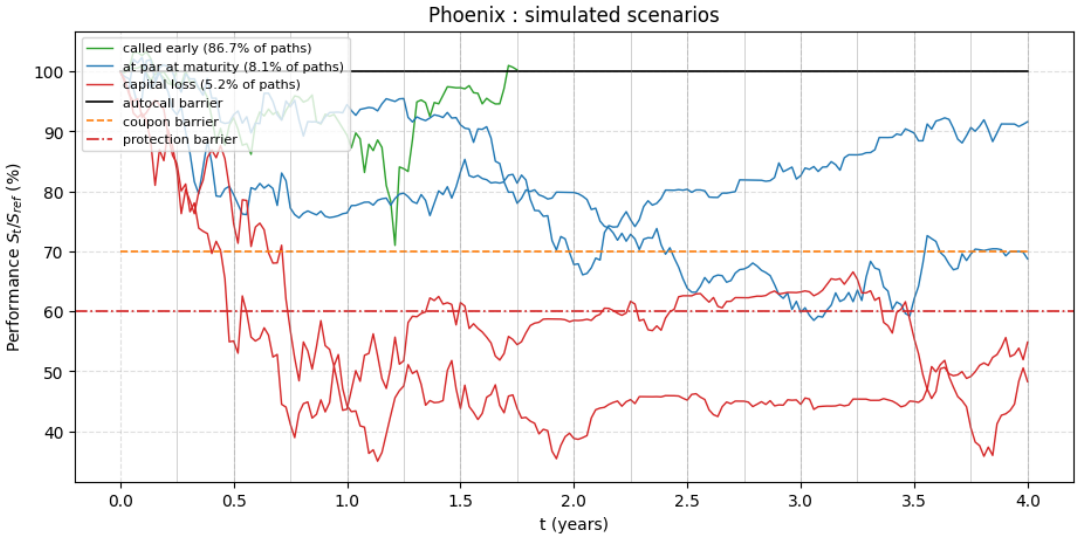


Figure 8: Base Phoenix: simulated paths of the performance, grouped by outcome.

Variant	Price at 6%	Fair coupon (% p.a.)	P(maturity) (%)	P(loss) (%)	E[loss] (%)	E[life] (y)	E[coupons]
Base (quarterly, memory, CB 70%)	99.96	6.04	13.29	5.22	3.05	1.03	3.28
No memory	99.67	6.41	13.29	5.22	3.05	1.03	3.28
Coupon barrier 50%	100.54	5.43	13.29	5.22	3.05	1.03	3.83
Coupon barrier 60%	100.30	5.67	13.29	5.22	3.05	1.03	3.61
Coupon barrier 80%	99.49	6.67	13.29	5.22	3.05	1.03	2.80
Coupon barrier = autocall barrier (= snowball Athena)	97.80	10.54	13.29	5.22	3.05	1.03	0.87
Unconditional coupon (CB 0%)	100.87	5.13	13.29	5.22	3.05	1.03	4.13
Monthly observations	99.91	6.19	7.77	3.13	1.83	0.59	5.53
Semi-annual observations	100.22	5.82	18.61	6.97	4.08	1.48	2.38
Annual observations	100.62	5.63	25.70	9.06	5.27	2.06	1.67
Autocall barrier 90%	100.19	5.46	3.13	1.70	1.01	0.42	1.37
Step-down 1.25% per quarter	99.86	6.20	7.14	4.29	2.54	0.84	2.62
Non-call 1 year	100.86	5.46	22.06	8.42	4.92	1.89	6.21
Protection 50%	100.71	5.16	13.29	3.43	2.25	1.03	3.28
Protection 70% (= coupon barrier)	99.28	6.85	13.29	7.32	3.78	1.03	3.28
American protection (continuous)	99.00	7.18	13.29	9.22	4.11	1.03	3.28
American protection (daily)	99.07	7.11	13.23	8.98	4.06	1.03	3.28
Airbag	101.28	4.49	13.29	5.22	1.61	1.03	3.28
Capital guaranteed 90%	102.26	3.33	13.29	5.22	0.52	1.03	3.28

Table 7: Phoenix variants on the CAC40 (Heston, 50,000 paths, standard errors of the prices between 0.02 and 0.06). P(call t_1) is 55.4% for every quarterly variant except the non-call period (0%) and the 90% autocall barrier (90.0%).

- *Observation frequency and early calls.* Contrary to the Athena, a Phoenix pays its coupons as long as it lives: everything that shortens its life shortens the coupon stream. Monthly observations raise the fair coupon (6.19%) and annual observations lower it (5.63%); a step-down of the autocall barrier raises it (6.20%), whereas it lowers the Athena coupon. A non-call period of one year lengthens the expected life to 1.89 years and lowers the fair coupon to 5.46%.
- *Protection.* The American protection costs 1.14 points of coupon (7.18%), the airbag gives back 1.55 points (4.49%), a 90% guarantee leaves 3.33%.

6.3 Greeks

The Phoenix is almost Delta-neutral at the money (0.047 for a 1% move against 0.41 for the Athena): a rise of the index makes the coupons more likely but also calls the note earlier, which stops a coupon stream priced above par. When the coupon is well above its fair level, the second

Phoenix	**Price**	**Delta** (per point)	**Delta 1%** (per 1% move)	**Vega** (per vol pt)	**Vanna** ($\times 10^4$)	**Rho** (per 1%)	**Dividend** (per 1%)	**Theta** (per day)
Base	99.96	0.00059	0.047	−0.335	6.13	−0.599	−0.325	0.0050
No memory	99.67	0.00106	0.085	−0.365	6.27	−0.560	−0.357	0.0053
American protection	99.00	0.00229	0.184	−0.394	5.72	−0.448	−0.437	0.0039
Airbag	101.28	−0.00151	−0.121	−0.224	5.38	−0.875	−0.101	0.0026
Unconditional coupon	100.87	−0.00077	−0.062	−0.253	5.67	−0.745	−0.202	0.0048

Table 8: Greeks of Phoenix variants (nominal 100, coupon 6%). Base case standard errors: Delta 0.00017, Vega 0.012, Vanna 0.8×10^{-4}, Rho 0.044, Dividend 0.044, Theta 0.0014; Gamma and Volga are not significant.

effect dominates and the Delta becomes negative (airbag and unconditional coupon, whose prices are 101.3 and 100.9). The Phoenix is short volatility (−0.33 per vol point) and short dividends (−0.32 per point, a third of the Athena): its coupons are paid on a shorter horizon, so the forward matters less. Figures 9 and 10 display the Greeks.

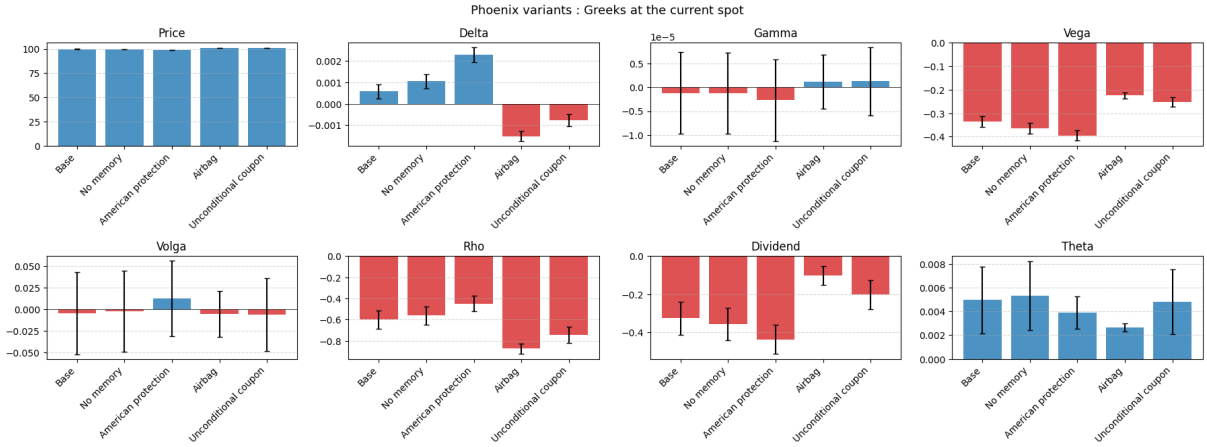


Figure 9: Greeks of the Phoenix variants with 95% confidence intervals.

Sensitivities to the Heston parameters. The base Phoenix has $\partial V/\partial v_0 = -40.4$, $\partial V/\partial\theta = -44.9$, $\partial V/\partial\kappa = -0.19$, $\partial V/\partial\xi = 1.36$ and $\partial V/\partial\rho = 0.87$: 58% to 65% of the Athena exposures to the variance level.

7 Athena versus Phoenix

The Athena concentrates its coupons in the scenarios where the index is up, which is why its fair coupon is higher and its Delta almost nine times larger; the Phoenix pays regularly as long as the index stays above 70%, a more stable income for a lower rate. With the same quarterly dates, the snowball Athena and the memory Phoenix with $B^{\text{cpn}} = B^{\text{call}}$ have the same fair coupon (10.54%), which is the identity of Section 2.

8 Implementation

8.1 Code

- `Produits/products.py`: `AutocallProduct` (common mechanics, `path_payoff`, `fair_coupon`, `statistics`), `Athena` and `Phoenix`;

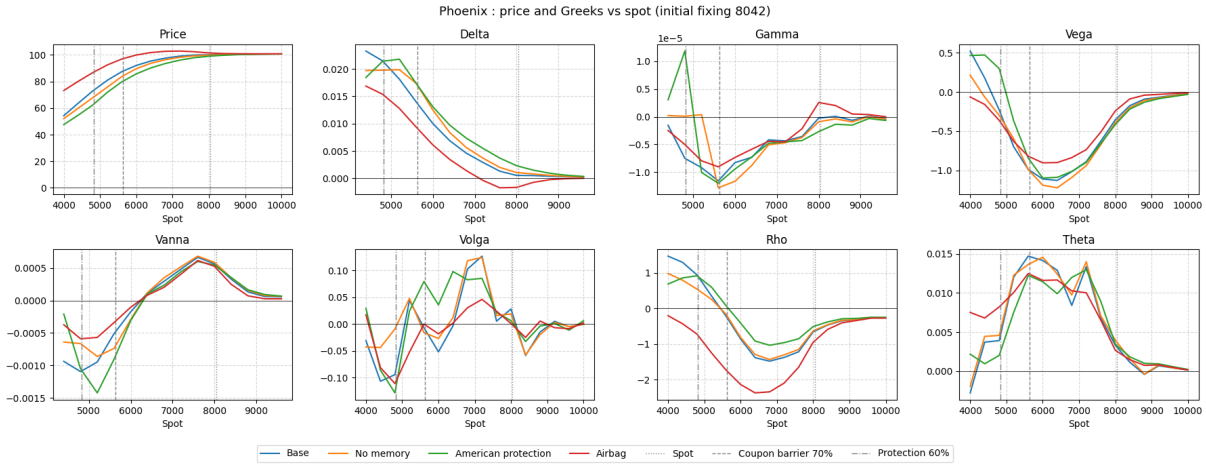


Figure 10: Phoenix: price and Greeks as functions of the spot (30,000 paths, spot step 400).

	Athena (annual, snowball)	**Phoenix** (quarterly, memory, CB 70%)
Fair coupon (% p.a.)	9.41	6.04
Probability of a call at the first date (%)	55.69	55.43
Expected life (years)	2.06	1.03
Probability of a capital loss (%)	9.06	5.22
Expected loss (% of nominal)	5.27	3.05
Delta (per 1% move)	0.407	0.047
Vega (per vol point)	−0.549	−0.335
Dividend (per 1% of carry)	−1.115	−0.325

Table 9: Base cases compared (European protection at 60%, autocall at 100%, 4 years).

- `Model/monte_carlo.py`: `MonteCarloPricer`, which simulates the paths and calls `path_payoff` for products with several cash flows;
- `Model/Heston.py`, `Model/forward_curve.py`: calibrated model and forward curve;
- `Model/greeks.py`: Greeks with common random numbers, including the Dividend Greek;
- `Produits/notes_analysis.py`: comparison of variants and plots;
- notebooks `Produits/Athena/athena_pricer.ipynb` and `Produits/Phoenix/phoenix_pricer.ipynb`.

```
from market import cac40_heston
from products import Athena, Phoenix
from monte_carlo import MonteCarloPricer
from greeks import Greeks

mkt = cac40_heston()
pricer = MonteCarloPricer(n_paths=50_000, steps_per_year=52, seed=42)
note = Phoenix(4.0, mkt.S0, 0.06, coupon_barrier=0.7, memory=True, frequency
    =0.25,
               autocall_barrier=1.0, protection_barrier=0.6, protection_type="
                   european")
print(pricer.price(mkt.heston, note))              # price and standard error
print(note.fair_coupon(pricer, mkt.heston))        # coupon p.a. at par
```

```
print(note.statistics(pricer, mkt.heston))          # call probabilities, loss,
    life, coupons
print(Greeks(pricer).compute(mkt.heston, note, Greeks.ORDER))
```

8.2 Cost

With N paths and M steps, the simulation costs $\mathcal{O}(NM)$ in time and stores S and v, i.e. $16(M+1)N$ bytes: 167 MB for $N = 50{,}000$ and the weekly grid of 4 years, 807 MB on a daily grid. Following the note through its n observation dates costs $\mathcal{O}(nN)$, the American continuous protection $\mathcal{O}(NM)$. Measured times (one CPU core): 0.30 s for a valuation including the simulation; 0.02 to 0.12 s for the price, the fair coupon and the statistics of a variant on cached paths (1.5 s with an American protection); 26 s for the eight Greeks of five notes (14 valuations each); 41 to 44 s for the spot profiles of four notes (16 spots $\times$ 6 scenarios, 30,000 paths).

9 Limitations and Next Steps

- *Single underlying.* Most autocalls sold today are worst-of on two to four indices or stocks; they need a multi-asset model and a correlation calibration.
- *Model.* The value of an autocall depends on the forward skew and on the joint dynamics of spot and volatility between observation dates; a pure Heston model with constant parameters is not the market standard, which is a local stochastic volatility model calibrated to the whole surface [5].
- *Fair coupon.* It prices the note at par without issuer margin, funding spread or issuer credit risk; a real issue price is below par by the margin.
- *Discretisation.* Euler full truncation with the Feller condition violated ($2\kappa\theta/\xi^2 = 0.49$), weekly grid with the variance frozen over each week for the Brownian bridge.
- *Greeks.* Gamma and Volga are not significant with 50,000 paths: the payoff jumps at the autocall and protection barriers.
- *Dividends.* Continuous dividend yield bootstrapped from the forwards; real notes are exposed to discrete dividends and to dividend risk, which the Dividend Greek only measures for a parallel shift.

10 Conclusion

A single parametrisation covers the Athena and the Phoenix with all their usual features, and a Monte Carlo pricer under a calibrated Heston model recovers five exact limit cases within one standard error. On the CAC40 of 12 February 2025, the base 4-year Athena pays 9.41% a year and the base Phoenix 6.04%; the protection type is the most expensive feature for the investor (1.77 point of coupon for an American protection on the Athena); more frequent observations raise the fair coupon of both products, whereas a step-down of the autocall barrier lowers the Athena coupon and raises the Phoenix one. Both notes are short volatility and short dividends, the Athena about three times more on dividends.

A Notation

Symbol	Meaning
N, S_{ref}, T	Nominal, initial fixing, maturity
t_k, $\delta_k = t_k - t_{k-1}$	Observation dates and accrual periods
$P_k = S_{t_k}/S_{\text{ref}}$	Performance at t_k
B_k^{call}, B_k^{cpn}, B^{prot}	Autocall, coupon and protection barriers (fractions of S_{ref})
c, $c^\star$	Coupon rate per annum, fair coupon
K_{put}, ϕ	Strike of the short put, capital floor
τ	Call date
$\mathbf{1}_{\text{KI}}$	Knock-in event of the protection

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